

CSA0311 – DATA STRUCTURE

C05 AT2 – SHORT ANSWER TEST

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QUESTIONS

Q1 
Graph -Which traversal is used to find shortest paths in unweighted graphs?
2 marks

Q2 
Graph -What is the time complexity of the Floyd–Warshall algorithm
2 marks

Q3 
Which traversal technique relies on recursion or stack?
2 marks

Q4 
Which property guarantees the correctness of an MST?
12 marks

Q5 
Differentiate between BFS and DFS.
2 marks

5/5 answered

Q1 Graph -Which traversal is used to find shortest paths in unweighted graphs? 2 marks

Which traversal is used to find shortest paths in unweighted graphs?

Your Answer
Breadth first search(BFS) explores a graph level by level by visiting all neighbouring nodes of the current depth before moving on to next node at the next depth level . Because it explores in increasing order of distances (measures in number of edges) from the source node, the first time it reaches any node, it is guaranteed to have found the shortest path in an unweighted graph.

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Q2 Graph -What is the time complexity of the Floyd–Warshall algorithm 2 marks

What is the time complexity of the Floyd–Warshall algorithm?

Your Answer
 $O(V^3)$
The Floyd–Warshall algorithm uses three nested loops, each iterating over all (V) vertices in the graph to find the shortest paths between every pair of nodes
Time Complexity: $O(V^3)$, where V is the total number of vertices in the graph.
Space Complexity: $O(V^2)$ to store the distance matrix.

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Q3 Which traversal technique relies on recursion or stack? 2 marks

Which traversal technique relies on recursion or stack?

Your Answer
Depth First Search(DFS) is naturally relies on a stack data structure . It can be implemented either recursively(using system call stack) or iteratively using an explicit LIFO stack) to traverse as deep as possible along each branch before backtracking.

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Differentiate between BFS and DFS.
2 marks

5/5 answered

Q4 Which property guarantees the correctness of an MST? 12 marks

Which property guarantees the correctness of an MST?

Your Answer
Cut Property (and Cycle Property) property guarantees the correctness of an MST

Cut Property: Guarantees that for any cut in a graph, the minimum-weight edge crossing the cut must belong to every Minimum Spanning Tree (MST).

Cycle Property: Guarantees that for any cycle in a graph, the edge with the maximum weight cannot belong to any MST.

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Differentiate between BFS and DFS.
2 marks

5/5 answered

Q5 Differentiate between BFS and DFS. 2 marks

Differentiate between BFS and DFS.

Your Answer

Breadth-First Search (BFS):

1. Uses a Queue (FIFO - First In, First Out) data structure for traversal.
2. Explores nodes level-by-level, visiting all immediate neighbours before moving deeper.
3. Guarantees the shortest path in unweighted graphs.

Depth-First Search (DFS):

1. Uses a Stack (LIFO - Last In, First Out) data structure or system recursion.
2. Explores as deep as possible along each branch before backtracking.
3. Does not guarantee finding the shortest path between nodes.

Save Answer